Multiple Linear Regression – Multi-Channel Marketing Analysis

Introduction

This project applies **Multiple Linear Regression (MLR)** to evaluate how different marketing channels — **TV, Radio, and Social Media** — collectively influence **Sales**. Unlike simple regression, which examines the effect of a single predictor, MLR allows us to model **multiple independent variables simultaneously**, providing a more realistic view of how marketing investments interact to drive outcomes.

By building an OLS regression model with all three predictors, checking for multicollinearity, validating assumptions, and interpreting coefficients, this analysis delivers **evidence-based recommendations** for marketing budget allocation. The goal is to help stakeholders understand not just *which* channel matters, but *how much each channel contributes when considered together*.

Key Points from Your Notebooks

1. Data Exploration

- Loaded marketing dataset with predictors: **TV, Radio, Social Media** and target: **Sales**.
- Performed exploratory data analysis (EDA): summary statistics, scatterplots, and correlation checks.

2. Multicollinearity Check

- Built a **correlation matrix** to see relationships among predictors.
- Calculated **Variance Inflation Factor (VIF)** to detect multicollinearity.
- Found that predictors were sufficiently independent for inclusion.

3. Model Building (OLS Regression)

- Fit an **Ordinary Least Squares (OLS)** regression model:
  [Sales \sim TV + Radio + SocialMedia]
- Reported **Adjusted R^2**, **F-statistic**, and **p-values** for each predictor.
- Iteratively dropped insignificant predictors (stepwise regression) to refine the model.

4. Assumption Validation

- Checked regression assumptions using diagnostic plots:
  - **Linearity**: Residuals vs. fitted values.
  - **Normality**: Q-Q plot and histogram of residuals.
  - **Homoscedasticity**: Residuals spread evenly across fitted values.
  - **Independence**: Durbin-Watson statistic confirmed residual independence.

5. Coefficient Interpretation

- **TV**: Positive, significant effect — each $1K spent increases Sales by a measurable amount.
- **Radio**: Strongest ROI per $1K spent — most efficient driver of Sales.
- **Social Media**: Not statistically significant in this dataset — limited direct impact when controlling for TV and Radio.

6. Business Recommendation

- **Prioritize Radio**: Highest ROI, most efficient channel.
- **Maintain TV**: Reliable contributor to Sales growth.
- **Reduce Social Media spend**: Reallocate toward Radio and TV until stronger evidence of impact emerges.
- **Strategic roadmap**: Short-term reallocation, medium-term reassessment, long-term regression-based optimization.

✅ This summary captures the **full lifecycle of your project**: feature selection, OLS modeling, assumption checking, coefficient interpretation, and business recommendations.

Would you like me to also condense this into a **one-slide executive summary** (bullet points only) so you can present it quickly to stakeholders without the technical detail?